

Relational Algebra, SQL & Normalization 1NF to BCNF

Subject: CS402 | Unit 3: Schema Design

1. Executive Overview & Syllabus Scope

Master normalization with real-world examples, functional dependency closure calculation, lossless join decomposition, and dep

2. Key Architectural Invariants & Core Topics

â€¢ Lecture Modules & Key Topics

- Unit 3: Relational Database Normalization Master Guide
- 1. Functional Dependencies: Armstrong's Axioms Reflexivity, Augmentation, Transitivity
- 2. First Normal Form 1NF: Eliminating repeating groups and atomic domain enforcement
- 3. Second Normal Form 2NF: Eliminating partial functional dependencies on candidate keys
- 4. Third Normal Form 3NF: Eliminating transitive dependencies $X \rightarrow Y$ where Y not in candidate key
- 5. Boyce-Codd Normal Form BCNF: Strict condition that for every $X \rightarrow Y$, X must be a superkey
- 6. Step-by-step algorithms for Canonical Cover and Lossless Join Decomposition

3. University Exam Tips & High-Yield Checklist

- * Practice numerical problems and time complexity derivations under timed conditions.
- * Draw neat labeled block diagrams for full credit in subjective university evaluations.
- * Use AI_Book Quiz & Summary features to test active recall before mid-terms and finals.